

1. Let's look at analyzing a real data set and acquiring all the information we need to perform a hypothesis test. Please describe the data, your analysis, and your findings using **complete sentences**. Use the data set named finches found at the underlined link. All units are in millimeters. You may want to reference the Lecture 7 examples file for an easier time loading in the data set.
 - (a) Make **and describe** an appropriate plot for the "beakheight" variable. In your description, include any observations you make regarding the **skew** (if there is any) and possible **outliers**.
 - (b) Calculate and interpret a 98% confidence interval for the numerical variable above. Within your code, define and display the variables necessary to calculate the confidence interval.
 - (c) Say you set up a hypothesis test for whether the mean beak height of the sample of finches is **different** from 13.5 mm. List the corresponding null and alternative hypothesis, using appropriate symbols for the parameter in question.
 - (d) Evaluate your hypothesis test using a level of confidence $\alpha = 0.02$.
 - (e) Does your confidence interval match the conclusion from your hypothesis test?
 - (f) Use RStudio to look at the 5th, 35th, and 55th observations in the data set (bracket notation is helpful for this). Which of these three birds has the tallest beak height?
 - (g) What would be the advantage(s) of using median instead of mean as a descriptor for beak height? Justify your answer based on the properties of the mean and median (you do not need to do a hypothesis test with the median).
2. A biologist at Chico State has calculated a 99% confidence interval for the mean weight in grams of the local population of western gray squirrels. This confidence interval is based on a simple random sample of 85 observations, and is given as (493, 639).

Calculate the **sample mean** and **standard deviation** by working through the calculations in reverse. Use the t -distribution in any calculations. You may either list your calculations in R or insert an image with your written work.